

Gases & Stoichiometry

Chapter 12

**Recall, that if we combine
Avogadro's theory with the
combined gas law, we can
create...**

the IDEAL Gas Law

$$PV=nRT$$

Ideal Gas Law

How did we get there??

- Recall: (for constant T & P)

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

- And (for a fixed amount of gas)

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

- Therefore

$$\frac{P_1 V_1}{n_1 T_1} = \frac{P_2 V_2}{n_2 T_2}$$

- This means

$$\frac{PV}{nT} = R$$

Where: R = Universal Gas Law Constant

Ideal Gas Law

The value of R can be calculated using one mole of a gas at STP:

$$P = 101.3 \text{ kPa}$$

$$V = 22.4 \text{ L}$$

$$T = 273 \text{ K}$$

$$n = 1.00 \text{ mol}$$

$$\frac{PV}{nT} = R$$

$$R = 8.314 \frac{\text{kPa L}}{\text{mol K}}$$

R can also be expressed in other units:

$$R = 0.08206 \frac{\text{atm L}}{\text{mol K}}$$

$$R = 62.37 \frac{\text{mmHg L}}{\text{mol K}}$$

The Ideal Gas Law Equation can be rearranged as:

$$PV = nRT$$

Ideal Gas Law

Sample Problems:

1. The pressure gauge on a 15.0 L cylinder of oxygen gas has a reading of 2.70×10^3 kPa and a temperature of 22.5°C . What is the mass of oxygen in the cylinder?
2. What volume of carbon dioxide did a swimmer breathe out when he exhaled 10.5 g of CO_2 @ SATP into a flask to measure his lung capacity?

HW:

-p.487 # 12-14

-p.488 #1-5

$$PV=nRT$$

Pressure



Temperature



Number of moles



$$PV=nRT$$

Volume



Gas constant



Real Gas Behaviour (p.483)

There are several assumptions in KMT that allow us to imagine all gases behaving in an “ideal” way. ie. All gas molecules are the same.

- i. Gas molecules have zero volume
- ii. Forces of attraction are zero
- iii. Molecules move in straight lines
- iv. Collisions are completely elastic

At high pressures and low temperatures, there are problems with the assumptions:

- v. In very small volumes there will be many more interactions and collisions
- vi. The volume of the gas molecules is now a significant fraction of the container

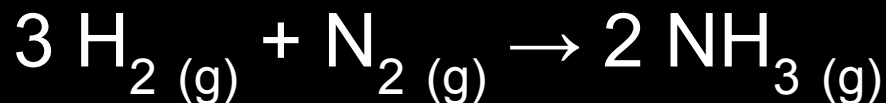
The Mole and Gases

- Many chemical reactions involve gases as reactants or products
- Measured at the same temperature and pressure, gases react or are chemically produced in **simple whole number volume ratios**. (Gay-Lussac's Law of Combining Gas Volumes)
- Coefficients for gaseous substances in a balanced equation represent **volume ratios** as well as **mole ratios**, if temperature and pressure conditions are the same. (Avogadro's Hypothesis)



1 mol 1 mol 2 mol

1 L 1 L 2 L



3 mol 1 mol 2 mol

3 L 1 L 2 L

12.3 Gas Law Stoichiometry

A balanced chemical equation can be used to solve problems for gases.

Sample Problems:

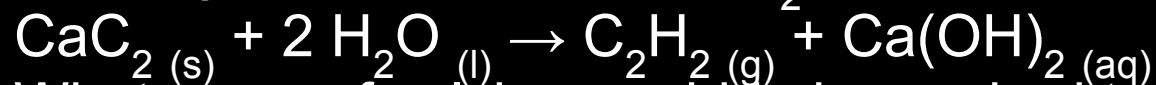


Suppose that in a reaction, 5.0 L of nitrogen reacts with hydrogen gas at the same T & P:

- a) What volume of ammonia is produced?
- b) What volume of hydrogen is consumed?

2. Zinc reacts with sulfuric acid to produce 380 mL of hydrogen gas measured at 101.6 kPa and 25°C. What mass of zinc is consumed in this reaction?

3. Acetylene gas, C_2H_2 , is normally prepared in the laboratory by reacting calcium carbide CaC_2 , with water.



What mass of calcium carbide is required to produce 235 mL of acetylene gas at SATP?